Vortrix Physics Quick Manual – Installation (Project Generation and Compilation)

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GitHub: https://github.com/Conqster/PreliminaryVortrixPhysics

Minimum Requirements

- OS: Windows x86/x64
- Processor (CPU): Intel 4th Generation Haswell / AMD Zen or newer (AVX2 and FMA instruction set support).
- Project Generation: CMake 3.18 or later
- Compiling: Visual Studio 2022+ (MSVC v143), C++ 17
- Build Configuration: Debug/Development/Distribution/Release/ReleaseASan
- Graphics: OpenGL 4.6 (hardware context)

Note: Older processors lacking AVX2 extensions can be targeted by explicitly changing the build option to "-DUSE_AVX2=OFF" during the command-line CMake generation step.

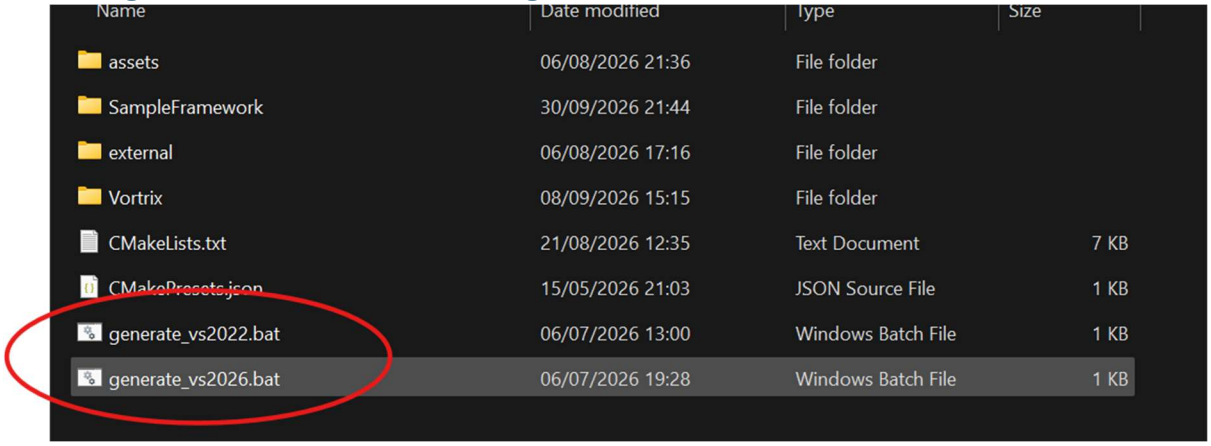
Recommended Requirements

The following setup was used for the dissertation benchmarks and performance analysis:

- Window x64
- Processor (CPU): AMD Ryzen 7 6800H (Primary) / Intel Core i5-10400F
- CMake 3.18
- Visual Studio 2026+ (MSVC v14.50), C++ 17
- Development (Optimised execution speeds with symbol maps enabled)

The provided executable, "Vortrix_PhysicsEngine/VortrixPhysicsSample.exe", was built using the recommended configuration described above.

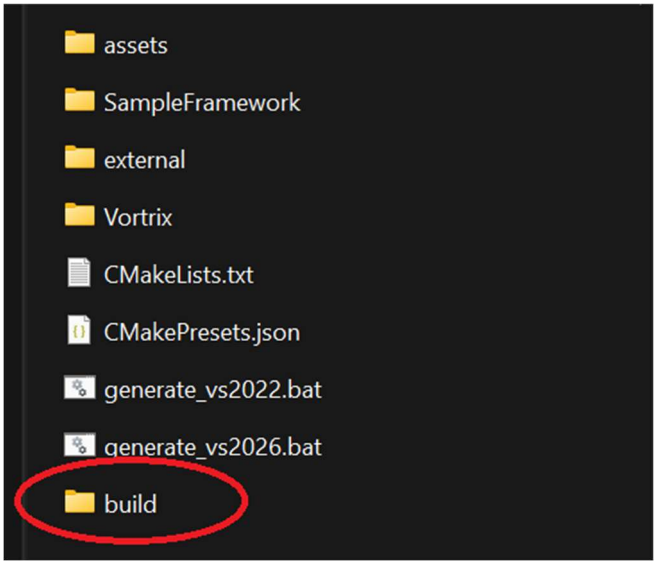
Building the Source Code Using the Provided Batch Files

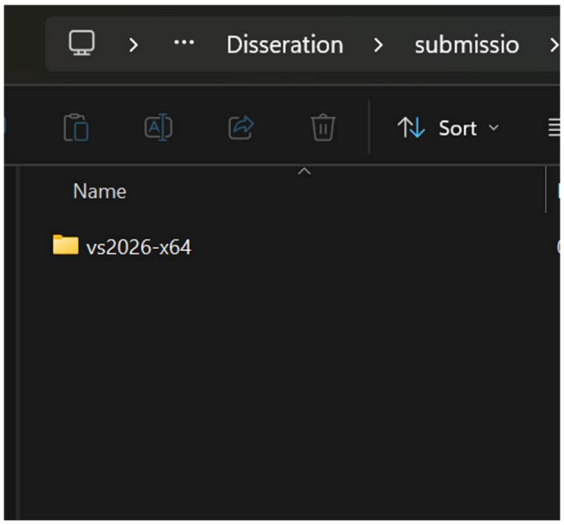


Two bat files are created for easy project generation:

- generate_vx2022: generates a Visual Studio 2022 project
- generate_vx2026: generates a Visual Studio 2026 project

Running either of the batch files generates the corresponding build directory: “build/vs2022-x64” and “build/vs2026-x64”.





After running the appropriate batch file, open the corresponding build directory and open the generated Visual Studio solution: “VortrixPhysics.slnx” or, for Visual Studio versions that use the traditional solution format: “VortrixPhysics.sln”.

Building the Source Code Using CMake Command Line

The Visual Studio 2026 project can also be generated directly from the command line:

```
cmake -S . -B build -G "Visual Studio 18 2026" -A x64
```

- S: specifies the source directory.
- B: specifies the build directory.

Specifying a separate build directory is recommended because it keeps generated files separate from the source code and makes it easier to clean the project.

The generated project can then be opened from:

“build“ or the specified build directory.

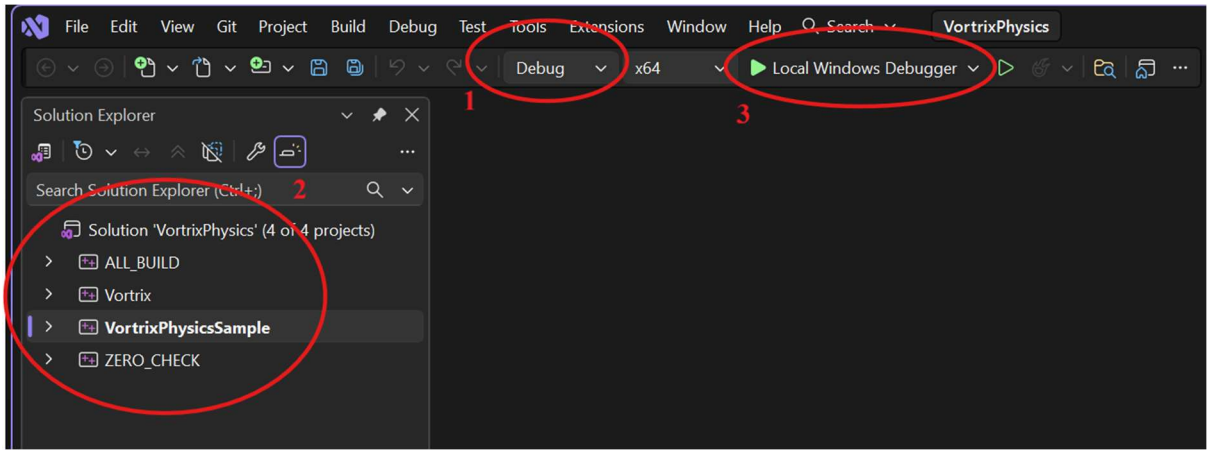
To disable AVX2 instruction set;

```
cmake -S . -B build -G "Visual Studio 18 2026" -A x64 -DUSE_AVX2=OFF
```

Code Compilation

When using the generated Visual Studio project, the default build configuration is **Debug**. The build configuration can be changed within Visual Studio to one of the configurations supported by the project:

- Debug
- Development
- Distribution
- Release
- ReleaseASan



1. Select Build Configuration
2. Ensure “VortrixPhysicsSample” is highlighted in bold; if not right-click and select “set as Startup Project”
3. Run “Local Windows Debugger” or Press “F5” to build and run the program.

Alternatively, a configuration can be selected from the command line

```
cmake --build build --config Development
```

Highlighted in red, **build** is the relative path to the generated build project. If the command is executed from within the build directory, “.”, can be used instead:

```
cmake --build . --config Development
```

After compilation, the generated executable can be found in the corresponding configuration directory, for example “build/Development/VortrixPhysicsSample.exe”.